P1. Solution: (a) $A \rightarrow \tilde{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ $\quad \tilde{A}\tilde{A}^* = \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{array}{l} \lambda_1 = 3+\sqrt{5} \\ \lambda_2 = 3-\sqrt{5} \end{array}$

$\Rightarrow \sigma_1 = \sqrt{3+\sqrt{5}}, \quad \sigma_2 = \sqrt{3-\sqrt{5}}$

$\tilde{A} = \tilde{U}\tilde{\Sigma}\tilde{V}^* \Rightarrow \tilde{A}\tilde{A}^* = \tilde{U}\tilde{\Sigma}\tilde{\Sigma}^*\tilde{U}^* \Rightarrow \tilde{A}\tilde{A}^*\tilde{U} = \tilde{U}\tilde{\Sigma}\tilde{\Sigma}^*$

$\Rightarrow \tilde{A}\tilde{A}^* u_1 = \sigma_1^2 u_1 \Rightarrow u_1 = \begin{bmatrix} \sqrt{\frac{1}{10-4\sqrt{5}}} \\ \sqrt{\frac{9-4\sqrt{5}}{10-4\sqrt{5}}} \end{bmatrix}$

$\tilde{A}\tilde{A}^* u_2 = \sigma_2^2 u_2 \Rightarrow u_2 = \begin{bmatrix} -\sqrt{\frac{9-4\sqrt{5}}{10-4\sqrt{5}}} \\ \sqrt{\frac{1}{10-4\sqrt{5}}} \end{bmatrix}$

similarly.

$\tilde{A}^*\tilde{A} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$

$\tilde{A}^*\tilde{A} v_1 = \sigma_1^2 v_1 \Rightarrow v_1 = \begin{bmatrix} \sqrt{\frac{2}{5-\sqrt{5}}} \\ \sqrt{\frac{3-\sqrt{5}}{5-\sqrt{5}}} \end{bmatrix}$

$\tilde{A}^*\tilde{A} v_2 = \sigma_2^2 v_1 \Rightarrow v_2 = \begin{bmatrix} -\sqrt{\frac{3-\sqrt{5}}{5-\sqrt{5}}} \\ \sqrt{\frac{2}{5-\sqrt{5}}} \end{bmatrix}$

$\Rightarrow A = U\Sigma V^*$

where $U = \begin{bmatrix} \sqrt{\frac{1}{10-4\sqrt{5}}} & -\sqrt{\frac{9-4\sqrt{5}}{10-4\sqrt{5}}} & 0 \\ \sqrt{\frac{9-4\sqrt{5}}{10-4\sqrt{5}}} & \sqrt{\frac{1}{10-4\sqrt{5}}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\quad \Sigma = \begin{bmatrix} \sqrt{3+\sqrt{5}} & & \\ & \sqrt{3-\sqrt{5}} & \\ & & 0 \end{bmatrix}$

$V = \begin{bmatrix} \sqrt{\frac{2}{5-\sqrt{5}}} & -\sqrt{\frac{3-\sqrt{5}}{5-\sqrt{5}}} & 0 \\ \sqrt{\frac{3-\sqrt{5}}{5-\sqrt{5}}} & \sqrt{\frac{2}{5-\sqrt{5}}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b).

$\|A\|_1 = 2$

$\|A\|_2 = 3+\sqrt{5}$

$\|A\|_\infty = 3$

$\|A\|_F = \sqrt{6}$

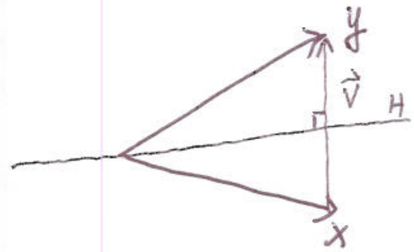
P2 (a). See Pages 51-52.

(b) See Page 52.

(c) see Page 58.

P3. Solution:

(a)



$\|x\|_2 = \|y\|_2 = 1$ (not necessarily 1)

$\vec{V} = y - x$

$$F = I - 2\frac{\vec{v}\vec{v}^*}{\vec{v}^*\vec{v}}$$
$$= I - 2\frac{(y-x)(y-x)^*}{(y-x)^*(y-x)}$$

(b) There doesn't exist such a F which is unitary to make $Fx = y$ when $\|x\|_2 \neq \|y\|_2$. Since $\|Fx\|_2 = \|x\|_2 \neq \|y\|_2$.

P4. Solution: $\det\begin{pmatrix} I & A \\ A^* & 0 \end{pmatrix} = \det\begin{pmatrix} I & A \\ 0 & -A^*A \end{pmatrix}$

$$= \det(I)\cdot\det(-A^*A) = (-1)^n \det(A^*A) \neq 0$$

$\Rightarrow \begin{bmatrix} I & A \\ A^* & 0 \end{bmatrix}\begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}$ has a unique solution.

$\begin{cases} r + Ax = b \\ A^* r = 0 \end{cases} \Rightarrow \begin{matrix} A^* r + A^*Ax = A^*b \\ A^* r = 0 \end{matrix} \Rightarrow A^*Ax = A^*b$

$\Rightarrow x = (A^*A)^{-1}A^*b$

i.e., the solution of the Least-squares problem.

P5. (a) See Pages 90-91, 103-104.

(b). See Pages 111.

P6. Solution: (a). We use induction.

i) $n=1$, $A = (a_{11})$, clearly $A = LU$ where $L = (1)$, $U = (a_{11})$

ii) Let us assume that the result holds when $n = m-1$.

$n=m$, $A = \begin{bmatrix} \tilde{A} & a_2 \\ a_1^T & a_{m\times m} \end{bmatrix}$ where $\tilde{A} \in \mathbb{C}^{(m-1)\times(m-1)}$, $a_1, a_2 \in \mathbb{C}^{m-1}$, $a_{m,m} \in \mathbb{C}$.

It is obvious that $\tilde{A}$ has a LU factorization, $\tilde{A} = \tilde{L}\tilde{U}$

$\left.\begin{matrix} \tilde{A} \text{ is nonsingular} \Rightarrow \tilde{U} \text{ is nonsingular} \\ \text{clearly, } \tilde{L} \text{ is nonsingular} \end{matrix}\right\} \Rightarrow \tilde{L}^{-1}, \tilde{U}^{-1}$ exist.

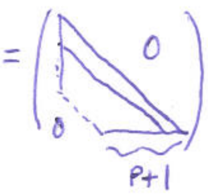
set $L = \begin{bmatrix} \tilde{L} & \\ a_1^T \tilde{U}^{-1} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} \tilde{U} & \tilde{L}^{-1} a_2 \\ & a_{mm} - a_1^T \tilde{U}^{-1} \tilde{L}^{-1} a_2 \end{bmatrix}$

Check

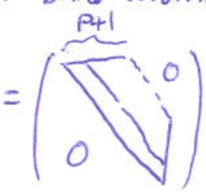
$$LU = \begin{bmatrix} \tilde{L}\tilde{U} & \tilde{L}\tilde{L}^{-1} a_2 \\ a_1^T \tilde{U}^{-1} \tilde{U} & a_1^T \tilde{U}^{-1} \tilde{L}^{-1} a_2 + a_{mm} - a_1^T \tilde{U}^{-1} \tilde{L}^{-1} a_2 \end{bmatrix} = \begin{bmatrix} \tilde{A} & a_2 \\ a_1^T & a_{mm} \end{bmatrix} = A.$$

is a LU factorization of A.

(b) L has a bandwidth $P+1$, i.e., $l_{ij} = 0$ for $i - j > P$

$L =$ 

U has a bandwidth $P+1$, i.e., $u_{ij} = 0$ for $j - i > P$.

$U =$ 

Extra Credit

Solution: A has a SVD factorization $A = U \Sigma V^*$

where $U \in \mathbb{C}^{m\times m}$, $\Sigma \in \mathbb{C}^{m\times n}$, $V \in \mathbb{C}^{n\times n}$, U, V unitary.

A is full rank $n \Rightarrow \Sigma = \begin{pmatrix} a_{11} & & \\ & \ddots & \\ & & a_{nn} \\ & 0 & \end{pmatrix}$ with $a_{ii} > 0$, $i = 1, \cdots, n$.

$$\begin{aligned} B &= A(A^*A)^{-1}A^* \\ &= U\Sigma V^* (V \Sigma^* U^* U \Sigma V^*)^{-1} V \Sigma^* U^* \\ &= U \Sigma (\Sigma^* \Sigma)^{-1} \Sigma^* U^* \\ &= U \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 0 \\ & & & & \ddots \\ & & & & & 0 \end{pmatrix} U^* \end{aligned}$$

$$\Rightarrow \|B\|_2 = \left\| \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 0 \\ & & & & \ddots \\ & & & & & 0 \end{pmatrix} \right\|_2 = 1$$